UNIVERSITY OF TRANSPORT HO CHI MINH CITY

INSTITUTE OF INFORMATION TECHNOLOGY



REQUIREMENT AND DESIGN DOCUMENT
STUDENT MANAGEMENT SYSTEM

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CHAPTER 1: FUNCTIONAL REQUIREMENTS

The Student Management System shall support academic and administrative operations for three user roles: Administrator, Student, and Lecturer. The system shall restrict access to features and data according to the role of each user

1. Authentication and Authorization

Registration: The system shall allow only the Administrator to create student and lecturer accounts based on official admission records or personnel information. The system shall require the following fields: automatically generated Student ID or Lecturer ID, Citizen ID Number, Full Name, Date of Birth, Email, Gender, Phone Number, and Faculty/Major. The system shall validate all inputs upon submission. The system shall reject the creation request and display an error message identifying the invalid field if any mandatory field is missing or any unique field already exists. Upon successful registration, the system shall automatically generate an account with default credentials and notify the user

Log in: The system shall allow registered users to select a login role (Administrator, Student or Lecturer) and log in using a username and password. After successful authentication, the system shall identify the user's role and display the interface and functions permitted for that role. If the credentials are invalid or the account is locked, the system shall reject the request and display an error message stating the reason

Log out: The system shall allow users to log out at any time. Upon logout, the system shall terminate the active session and clear all temporary session data

Password Change: The system shall allow students and lecturers to change their passwords by entering the current password and a new password. The system shall require the new password to meet the password complexity rules and to be confirmed by the user before saving the change

Password Recovery: The system shall allow users to recover their accounts by providing a registered email address or phone number. After successfully verifying account ownership, the system shall allow the user to set a new password

2. User Management

Student Management: The system shall allow the Administrator to create, view, update, and delete student records. The system shall require each student record to have an academic or administrative status. The system shall allow students to view and update their own contact information. The system shall not allow students to modify their identification data or affiliated faculty

Lecturer Management: The system shall allow the Administrator to create, view, update, and delete lecturer profiles and to assign lecturers to class sections. The system shall allow lecturers to view and update their own contact information. The system shall not allow lecturers to modify their identification data or affiliated faculty

3. Academic Management

Curriculum Management: The system shall allow students to view the standard curriculum of their major, including compulsory courses, elective courses, credit values, and prerequisites. The system shall automatically mark the completion status of each course based on the student's academic transcript

Semester Management: The system shall allow the Administrator to create and configure academic semesters, including start dates, end dates, course registration windows, and examination schedule timelines. The system shall link all academic activities and class section offerings to a specific semester

Faculty Management: The system shall allow the Administrator to create, view, update, and delete faculty records, each containing Faculty Code, Faculty Name, Email, and Phone Number. The system shall reject the deletion of any faculty that currently manages active courses or has enrolled students

Course Management: The system shall allow the Administrator to create, view, update, and delete courses. The system shall require each course record to include Course Code, Course Name, Credits, Managing Faculty, and Prerequisite Constraints. The system shall reject the deletion of a course that is linked to ongoing class sections or historical academic records

Class Section Management: The system shall allow the Administrator to manage multiple class sections of a course within a specific semester. The system shall require each class section record to include Class Code, Course Code, Semester, Assigned Lecturer, Maximum Capacity, Current Enrollment, Room, and Class Schedule

4. Grading and Attendance Management

Attendance Management: The system shall allow lecturers to record and update daily student attendance only for class sections assigned to them in the active semester. The system shall automatically calculate each student's attendance rate for each course based on the recorded presence and absence data

Gradebook Management: The system shall allow lecturers to record student grades for midterms, assignments, attendance, and final examinations. The system shall reject any numerical mark outside the range of 0.0 to 10.0. The system shall automatically calculate the final course grade using the weighted grading formula defined for each course and assign the corresponding letter grade

GPA Calculation: The system shall automatically calculate each student's Semester GPA and Cumulative GPA based on finalized academic results and credit-weighted grading rules

Academic Transcript Viewing: The system shall allow students to view their academic history, including component marks, final exam results, and course letter grades. The system shall automatically update the Semester GPA and Cumulative GPA. The system shall display only finalized grades to students

CHAPTER 2: NON-FUNCTIONAL REQUIREMENTS

Security: The system shall protect user data from unauthorized access. The system shall automatically terminate an active session after 15 minutes of inactivity, clear all session data, and redirect the user to the login page

Usability: The system shall provide a consistent and responsive web interface across all functions. The system shall display error messages that identify the invalid field or the schedule conflict that caused the error. The system shall display a confirmation or error message for every user action, including successful updates, failed operations, and invalid submissions

Reliability: The system shall maintain data integrity and consistency during database operations. The system shall confirm completion of an operation only after all related record changes have been committed successfully. If a database operation fails, the system shall roll back all related changes, display an error message, and continue running without crashing

Performance: The system shall be designed for small-scale institutional deployment. Under normal operating conditions, the system shall respond to user actions within 15 seconds

Maintainability: The system shall follow a modular architecture to support testing, debugging, and extension. The source code shall include comments, follow consistent naming conventions, and comply with the PEP 8 coding standard. The system structure shall allow business rules to be modified and the data storage mechanism to be replaced without redesigning the entire system

Design Constraints: The system shall be a web-based application. The user interface shall be developed using HTML, CSS, and JavaScript. The core business logic shall be implemented in Python. The system shall use a MySQL Relational Database Management System (RDBMS) for data persistence. The system shall be deployed and hosted on a Windows 11 server environment

CHAPTER 3: DATA FLOW DIAGRAM

1. Overview of DFD

This document describes two levels of the Data Flow Diagram (DFD) of the Student Management System (SMS). A DFD is a modeling tool used to represent external entities, processing functions and the data flows exchanged between them

The two DFD levels included in this document are:

- **DFD Level 0 (Context Diagram):** A high-level overview that treats the entire system as a single process and shows its interactions with three external entities: Administrator, Student and Lecturer
- **DFD Level 1:** Decomposes the main process into four sub-processes, showing the detailed data flows between these sub-processes and the external entities, and the data passed between sub-processes

2. DFD Diagram Level 0 - Context Diagram

2.1. Overview

DFD Level 0, also known as the Context Diagram, represents the entire Student Management System as a single process labeled “0 - Student Management System”. The diagram defines the system boundary and shows the data flows exchanged between this process and three external entities: Administrator, Student and Lecturer. No internal processing is shown at this level

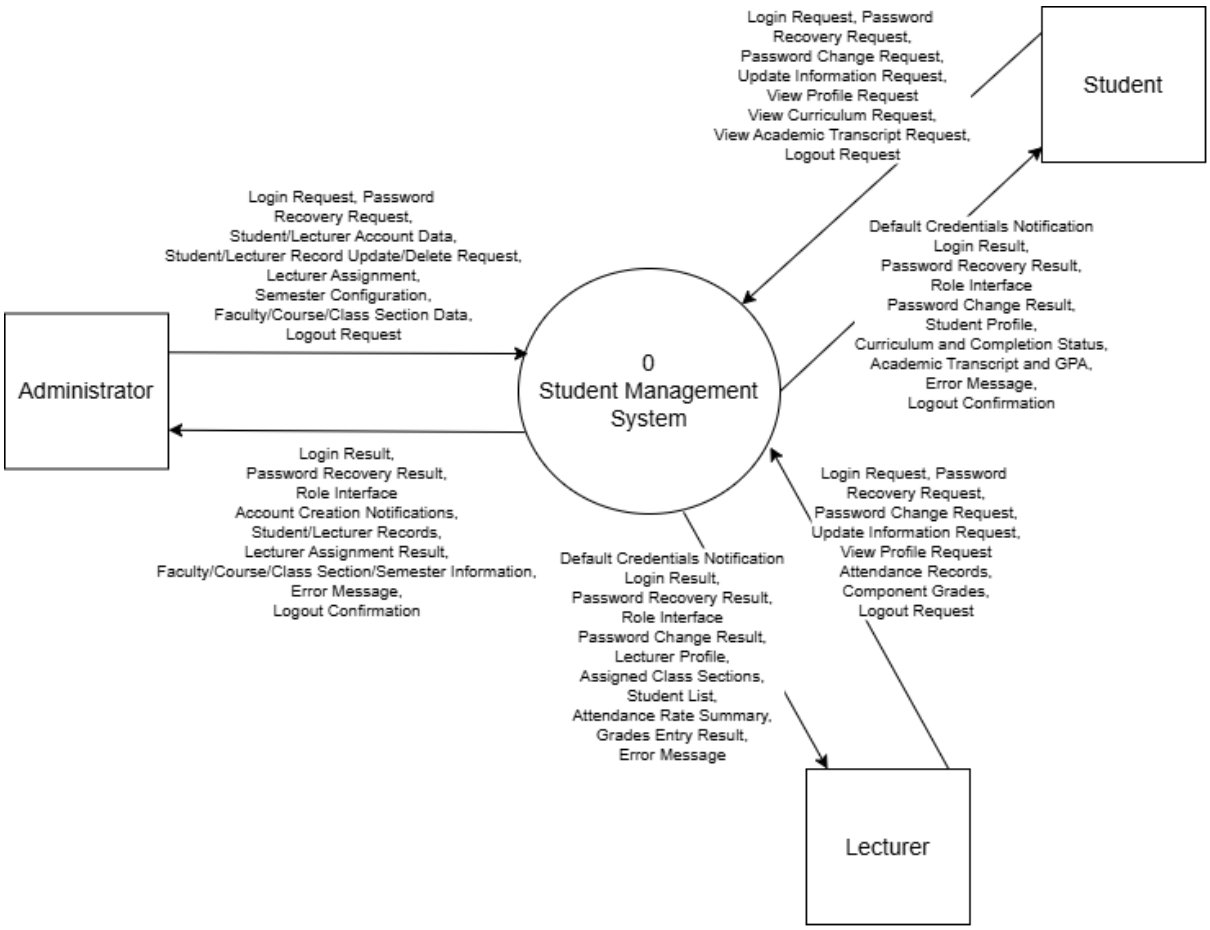


Image 1: DFD Diagram Level 0

2.2. External Entities

Entity	Role	Description
Administrator	System Administrator	Manages the entire system with the highest level of access, responsible for user management, system configuration and academic data management.
Student	End User	Uses the system to view academic transcripts and GPA, view the curriculum, check completion status, view and update personal information.
Lecturer	End User	Uses the system to record attendance, enter component grades, view assigned class sections,

		student lists and attendance rate summaries, view and update personal information.

3. DFD Diagram Level 1

3.1. Overview

DFD Level 1 decomposes the central process “0 - Student Management System” into four sub-processes, showing the detailed data flows between these sub-processes and the three external entities (Administrator, Student and Lecturer), as well as the data passed between the sub-processes themselves

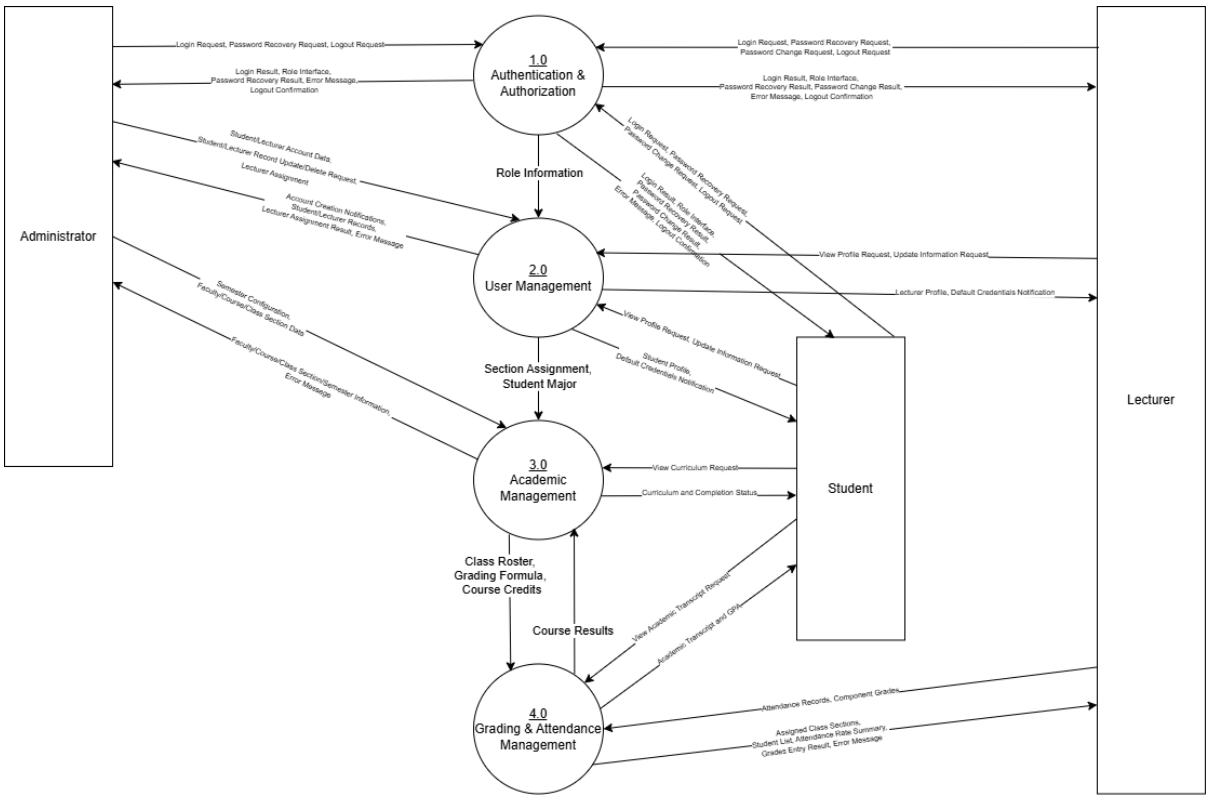


Image 2: DFD Diagram Level 1

3.2. Processes

Process ID	Process Name	Functional Description

1.0	Authentication & Authorization	Handles identity verification and role-based access control for all users (Administrator, Student, Lecturer). Manages login, password recovery, password change and logout
2.0	User Management	Manages student and lecturer accounts and records, including creation, update and deletion. Handles lecturer assignment and personal contact information updates
3.0	Academic Management	Manages academic data including semester configuration, faculty, course, curriculum and class section data. Only the Administrator can configure this data
4.0	Grading & Attendance Management	Manages attendance records and component grade entry by Lecturers. Calculates course results and GPA for Students

CHAPTER 4: USE-CASE DIAGRAM

1. Use-Case Diagram



Image 3: Use-Case Diagram

1.1. List of Actor

No	Actor	Description
1	User	Represents an individual who uses the system's authentication functions, including selecting a login role (Administrator, Student or Lecturer), logging in and recovering a password. After successful login, acts as an Administrator, Student or Lecturer according to the role stored in the account
2	Student	Represents a student who can change password, view and update personal information, view the curriculum, view academic transcript and log out
3	Lecturer	Represents a lecturer who can change password, view and update personal information, manage attendance, manage gradebooks for assigned class sections and log out
4	Administrator	Represents the person responsible for creating accounts, managing student and lecturer records, assigning lecturers to class sections, managing academic arrangements: semesters, faculties, courses, class sections and logging out

1.2. List of Use-Case

No	Use Case	Description
1	Log In	Allows registered users to select a login role and access the system with their

		credentials, then opens the functions permitted for their role
2	Recover Password	Allows users to set a new password after verifying ownership through the registered email address or phone number
3	Change Password	Allows students and lecturers to replace the current password with a new one that meets the complexity rules
4	View Profile	Allows students and lecturers to view their personal and contact information
5	Update Information	Allows students and lecturers to update their own contact information. Identification data and faculty cannot be modified
6	Log Out	Allows users to log out at any time. The system terminates the active session and clears all temporary session data
7	View Curriculum	Allows students to view the curriculum of their major and the completion status of each course

8	View Academic Transcript	Allows students to view their finalized grades, Semester GPA and Cumulative GPA
9	Manage Attendance	Allows lecturers to record and update attendance for their assigned class sections in the active semester
10	Manage Gradebook	Allows lecturers to record student marks. The system validates the marks and calculates the final course grade
11	Create Account	Allows only the administrator to create student and lecturer accounts. The system validates the input and notifies the user
12	Manage Student Records	Allows the administrator to create, view, update and delete student records
13	Manage Lecturer Profiles	Allows the administrator to create, view, update, and delete lecturer profiles
14	Assign Lecturer to Class Section	Allows the administrator to assign a lecturer to a class section
15	Manage Semesters	Allows the administrator to configure semesters, registration windows and examination schedules

16	Manage Faculties	Allows the administrator to manage faculty records. A faculty with active courses or enrolled students cannot be deleted
17	Manage Courses	Allows the administrator to manage course records. A course linked to class sections or academic records cannot be deleted
18	Manage Class Sections	Allows the administrator to manage the class sections of a course in a semester

2. Use-Case Log In

2.1. Summary

Allows registered users to select a login role (Administrator, Student or Lecturer) and log in to the system using a username and password

2.2 Flow of Event

2.2.1. Main Flow

1. The user opens the login page and selects the login role: Administrator, Student or Lecturer
2. The user enters the username and password, then submits the login request
3. The system validates the credentials against the MySQL database
4. The system identifies the user role: Administrator, Student or Lecturer and checks that it matches the selected role
5. The system displays the interface and functions permitted for that role

2.2.2. Alternative Flow

A1: If the credentials are invalid or the account is locked, the system rejects the request and displays an appropriate error message

A2: At Step 4, if the role stored in the account differs from the selected role, the system rejects the request, creates no session, and displays the same message as for invalid credentials (“Invalid username or password.”). The system does not reveal the account’s actual role

2.3. Special Requirement

The system shall respond to the login request within 15 seconds under normal operating conditions, restrict access to features and data according to the role of each user, and automatically terminate an active session after 15 minutes of inactivity, clear all session data, and redirect the user to the login page

2.4. Precondition

The user has a valid registered account created by the Administrator with an assigned role

2.5. Postconditions

Success: The user is authenticated and an active session is initiated.

Failure: No session is created and an error message is displayed.

2.6. Extension Points

Recover Password

3. Use-Case Recover Password

3.1. Summary

Allows users to recover their accounts by setting a new password after verifying account ownership through a registered email address or phone number

3.2 Flow of Event

3.2.1. Main Flow

1. The user selects the ‘’Recover Password” option on the login page
2. The user provides the registered email address or phone number and submits the recovery request

3. The system verifies the provided email address or phone number against the MySQL database
4. Once account ownership is verified, the system allows the user to set a new password
5. The user enters the new password and confirms it
6. The system validates the password complexity, updates the user's record in the MySQL database and displays a success message

3.2.2. Alternative Flow

If the email address or phone number is not registered, account ownership cannot be verified, or the new password does not meet the complexity rules, the system rejects the request and displays an error message

3.3. Special Requirement

The system shall respond to the password recovery request within 15 seconds under normal operating conditions, display clear error messages for invalid inputs, and confirm completion only after password updates are committed successfully to the database without exposing sensitive user credentials

3.4. Precondition

The user has a registered account in the system and cannot access their account or has requested a password reset from the Log In interface

3.5. Postconditions

Success: The new password is saved and the user can log in with it

Failure: The password remains unchanged and an error message is displayed

3.6. Extension Points

None

4. Use-Case Change Password

4.1. Summary

Allows students and lecturers to change their password by entering their current password and a new password meeting complexity rules

4.2 Flow of Event

4.2.1. Main Flow

1. The user selects the “Change Password” option
2. The system prompts the user to enter the current password, the new password, and the confirmation of the new password
3. The user enters the required information and submits the request
4. The system verifies the current password against the MySQL database
5. The system validates that the new password meets the password complexity rules and matches the confirmation
6. The system updates the user’s record in the MySQL database and displays a success message

4.2.2. Alternative Flow

If the current password is incorrect, the new password does not meet the complexity rules, or the new password confirmation does not match, the system rejects the request and displays an appropriate error message

4.3. Special Requirement

The system shall respond to the password change request within 15 seconds under normal operating conditions, display clear error messages for invalid inputs, and confirm completion only after password updates are committed successfully to the database without exposing sensitive user credentials

4.4. Precondition

The user is authenticated as a Student or Lecturer and has an active session in the system

4.5. Postconditions

Success: The new password is saved and replaces the current password

Failure: The password remains unchanged and an error message is displayed

4.6. Extension Points

None

5. Use-Case View Profile

5.1. Summary

Allows students and lecturers to view their personal and account profile information

5.2 Flow of Event

5.2.1. Main Flow

1. The user selects the “View Profile” option
2. The system retrieves the user’s personal information from the MySQL database
3. The system displays the user’s profile details on the interface

5.2.2. Alternative Flow

If the personal information cannot be retrieved, the system displays an error message

5.3. Special Requirement

The system shall respond to the request within 15 seconds under normal operating conditions. The system shall restrict access so that users can only view their own profile

5.4. Precondition

The user is authenticated as a Student or Lecturer and has an active session in the system

5.5. Postconditions

Success: The personal information of the user is displayed.

Failure: An error message is displayed

5.6. Extension Points

None

6. Use-Case Update Information

6.1. Summary

Allows students and lecturers to update their own contact information while preventing modifications to identification data or affiliated faculty

6.2 Flow of Event

6.2.1. Main Flow

1. The user selects the “Update Information” option
2. The system displays the current contact information of the user
3. The user edits the contact information and submits the request
4. The system validates the submitted information
5. The system updates the user’s record in the MySQL database and displays a success message

6.2.2. Alternative Flow

If the submitted information is invalid or the user attempts to modify identification data or affiliated faculty, the system rejects the request and displays an error message identifying the invalid field

6.3. Special Requirement

The system shall respond to the update request within 15 seconds under normal operating conditions. The system shall not allow users to modify their identification data or affiliated faculty and if the input format is invalid, the system displays an error message identifying the invalid field

6.4. Precondition

The user is authenticated as a Student or Lecturer and has an active session in the system

6.5. Postconditions

Success: The contact information of the user is updated.

Failure: The contact information remains unchanged and an error message is displayed

6.6. Extension Points

None

7. Use-Case View Curriculum

7.1. Summary

Allows students to view the standard curriculum of their major, including compulsory courses, elective courses, credit values, and prerequisites

7.2 Flow of Event

7.2.1. Main Flow

1. The student selects the “View Curriculum” option
2. The system retrieves the standard curriculum of the student’s major and the student’s academic transcript from the MySQL database
3. The system marks the completion status of each course based on the academic transcript
4. The system displays the compulsory courses, elective courses, credit values, prerequisites, and completion status of each course

7.2.2. Alternative Flow

If the curriculum or the academic transcript cannot be retrieved, the system displays an error message

7.3. Special Requirement

The system shall respond to the request within 15 seconds under normal operating conditions. The system shall automatically mark and update the completion status of

each course based on the student's academic transcript and restrict access so that students can only view the curriculum of their own major

7.4. Precondition

The user is authenticated as a Student and has an active session in the system

7.5. Postconditions

Success: The curriculum and the completion status of each course are displayed

Failure: An error message is displayed

7.6. Extension Points

None

8. View Academic Transcript

8.1. Summary

Allows a student to view their academic history and results.

8.2. Flow of Events

8.2.1. Main Flow

1. The use case starts when the student selects "Transcript".
2. The system retrieves the student's finalized results from the database.
3. The system displays component marks, final exam results, and letter grades per course.
4. The system displays Semester GPA and Cumulative GPA.

8.2.2. Alternative Flow

A1: At Step 2, if no finalized grades exist for the student: The system displays a "no data" message: "No finalized grade data available."

A2: At Step 2, if database access or verification fails due to technical errors: The system displays an error message: "System error: Failed to retrieve transcript data. Please try again later."

8.3. Special Requirements

Only finalized grades are shown; unfinalized/draft marks must not be displayed.

GPA must update automatically based on finalized grades.

Response time from selecting "Transcript" to rendering the full transcript must not exceed 15 seconds under standard test load conditions

8.4. Preconditions

The student is logged in.

8.5. Postconditions

The transcript is displayed; data is unchanged.

8.6. Extension Points

None

9. Log out

9.1. Summary

Allows a user to exit the system

9.2. Flow of Events

9.2.1. Main Flow

1. The use case starts when the user selects “Log out”.
2. The system terminates the active user session in the database
3. The system clears temporary session data.
4. Upon successful session invalidation in the database, the system redirects the user to the login page.

9.2.2. Alternative Flow

If no user activity is detected for 15 minutes: The system automatically performs Steps 2–4, invalidates the session, and redirects to the login page with a notification: "Session expired due to inactivity."

9.3. Special Requirements

Logout must be available at any time from any screen for logged-in users.

The login page redirection and logout success state must only occur after session deletion is confirmed in the database state.

All temporary session data and cached sensitive data must be completely cleared.

9.4. Preconditions

The user is logged in.

9.5. Postconditions

Success: Session terminated in database, temporary session data cleared, user redirected to login page.

Failure: Session remains active in the database; user context and local data remain unchanged; error message is displayed.

9.6. Extension Points

None

10. Manage Attendance

10.1. Summary

Allows a lecturer to record and update student attendance for assigned class sections.

10.2. Flow of Events

10.2.1. Main Flow

1. The use case starts when the lecturer selects “Attendance”.
2. The system retrieves and lists the lecturer’s assigned sections in the active semester from the database.
3. The lecturer selects a section and date.
4. The system retrieves and lists the lecturer’s assigned sections in the active semester from the database.

5. The lecturer marks attendance status for each student and clicks “Save”.
6. The system validates and saves the attendance record to the database, then recalculates student attendance rates.
7. Upon successful database update, the system displays a success message.

10.2.2. Alternative Flow

A1: At Step 2 or 4, if the lecturer profile or selected section does not exist in the database: The system rejects the request and displays an error message: "Section or lecturer information does not exist."

A2: At Step 6, if saving attendance data fails due to system error or verification/constraint failure: The system rolls back the transaction, maintains original data, and displays an error message: "Save failed: Unable to update attendance data. Transaction rolled back."

10.3. Special Requirements

Lecturers may only manage attendance for assigned sections within the active semester.

Attendance rates must be automatically recalculated upon saving.

Response time for saving attendance and updating database records must not exceed 15 seconds under standard test load conditions

10.4. Preconditions

The lecturer is logged in; an active semester exists; the lecturer is assigned to a section that has students.

10.5. Postconditions

Success: Attendance records and attendance rates are updated in the database.

Failure: Database remains unchanged.

10.6. Extension Points

None

11. Manage Gradebook

11.1. Summary

Allows a lecturer to record and update student grades for assigned class sections.

11.2. Flow of Events

11.2.1. Main Flow

1. The use case starts when the lecturer selects “Gradebook”.
2. The lecturer selects an assigned section.
3. The system verifies assignments and displays the list of students along with grade components (midterm, assignments, attendance, final exam).
4. The lecturer enters or updates component marks and clicks “Save”.
5. The system validates the entered marks
6. The system calculates the final grade using the course’s weighted grading formula, determines the letter grade, updates GPA, and saves all data to the database.
7. Upon successful database update, the system displays a success message.

11.2.2. Alternative Flow

A1: At Step 5, if any entered mark is outside the allowed range (0.0 – 10.0) or invalid: The system rejects the input and displays an error message identifying the invalid field: "Invalid grade format or out-of-range value at field [Field Name]. Grades must be between 0.0 and 10.0."

A2: At Step 6, if saving grade data to the database fails due to a system error or failed verification: The system rolls back the transaction, maintains original grades, and displays an error message: "Save failed: Unable to update gradebook data. Transaction rolled back."

11.3. Special Requirements

Final numerical grades, letter grades, and GPA impact must be calculated automatically; students can only view grades after they are finalized.

Response time for grade validation, formula calculation, and database persistence must not exceed 15 seconds under standard test load conditions

11.4. Preconditions

The lecturer is logged in and assigned to the class section; the course has a defined grading formula; the section has enrolled students.

11.5. Postconditions

Success: Component marks, final grade, letter grade, and GPA are saved in the database.

Failure: Grade records remain unchanged in the database.

11.6. Extension Points

None

12. Create Account

12.1. Summary

Allows the Administrator to create new student and lecturer accounts in the system.

12.2. Flow of Events

12.2.1. Main Flow

1. The use case starts when the Administrator selects “Create account”.
2. The Administrator selects the account type (student/lecturer).
3. The Administrator enters the required personal, identification, contact, and faculty or major information according to the selected account type
4. The system validates the submitted information, including mandatory fields and unique-field constraints
5. The system automatically generates the appropriate Student ID or Lecturer ID and creates the account with the initial credentials
6. Upon successful database commitment, the system notifies the Administrator with a success message and sends account details to the user.

12.2.2. Alternative Flow

A1: At Step 4, if mandatory fields are missing or unique constraints are violated (e.g., email/ID already exists): The system rejects the request and displays an error message naming the invalid field: "Validation error: [Field Name] is missing or already exists in the system."

A2: At Step 5, if saving the new account to the database fails due to technical errors or system constraints: The system rolls back the transaction, creates no account, and displays an error message: "System error: Unable to create account in database. Transaction rolled back."

12.3. Special Requirements

Only Administrators are authorized to create accounts, based on official admission or personnel records.

Response time for validating input, generating IDs, saving to database, and returning the success message must not exceed 15 seconds under standard test load conditions

12.4. Preconditions

The Administrator is logged in.

12.5. Postconditions

Success: New account, ID, and profile are stored in the database, and notification is sent.

Failure: No account or profile is created; database remains unchanged.

12.6. Extension Points

None

13. Manage Student Records

13.1. Summary

Allows the Administrator to create, view, update, and delete student records, while students can view and update their own contact information.

13.2. Flow of Events

13.2.1. Main Flow

1. The use case starts when the Administrator selects “Student management”.
2. The system displays the student list (searchable).
3. The Administrator chooses to add, view, update, or delete.
4. For Add/update: enter profile data and academic/administrative status; the system validates and saves.
5. For Delete: The system checks whether the student record can be deleted without violating data integrity constraints. If deletion is allowed, the system asks the Administrator for confirmation. After confirmation, the system deletes the record and displays the result.
6. Upon successful database commitment, the system displays a success message with the operation result.

13.2.2. Alternative Flow

A1: Student self-update:

1. The student opens their own profile.
2. The system displays it; only contact fields (email, phone, etc.) are editable.
3. The student edits contact details and clicks Save.
4. If the student tries to modify identification data or affiliated faculty: "Unauthorized operation or invalid input: Students may only modify contact information."

A2. At Step 4, if a mandatory field is missing or a unique field is duplicated: the system rejects the request and displays “Validation error: [Field Name] is missing or already exists.”

A3. At Step 5, if deletion violates data integrity constraints: the system rejects the request and displays the reason.

A4. At Step 5, if the Administrator cancels the confirmation: no record is deleted.

13.3. Special Requirements

Each student record must explicitly include an academic or administrative status. Students can only access their own records and modify contact details; identification data and affiliated faculty must be read-only for students.

Response time for validating, saving, or deleting records in the database must not exceed 15 seconds under standard test load conditions

13.4. Preconditions

The user is logged in.

13.5. Postconditions

Success: Student record is created, updated, or deleted in the database.

Failure: Student database records remain unchanged; error message is displayed.

13.6. Extension Points

None

14. Manage Lecturer Profiles

14.1. Summary

Allows the Administrator to create, view, update, and delete lecturer profiles, while lecturers can view and update their own contact information.

14.2. Flow of Events

14.2.1. Main Flow

1. The use case starts when the Administrator selects “Lecturer management”.
2. The system displays the lecturer list (searchable).
3. The Administrator chooses to add, view, update, or delete.
4. Add/update: enter lecturer profile data; the system validates and saves.

5. Delete: The system checks whether the lecturer profile can be deleted without violating data integrity constraints. If deletion is allowed, the system asks the Administrator for confirmation. After confirmation, the system deletes the profile and displays the result
6. Upon successful database commitment, the system displays a success message.

14.2.2. Alternative Flow

A1: Lecturer self-update:

1. The lecturer opens their own profile.
2. The system displays it; only contact fields are editable.
3. The lecturer edits contact details and clicks Save.
4. The system validates and updates the database.
5. If the lecturer attempts to modify identification data or affiliated faculty: the system rejects the update, performs rollback, and displays an error message: "Unauthorized operation: Lecturers may only modify contact information."

A2: At Step 4, if required fields are missing or unique constraints are violated: The system rejects the request, performs rollback, and displays an error message naming the invalid field: "Validation error at field [Field Name]: Invalid data or duplicate value."

A3: At Step 5, if confirmation is cancelled by the Administrator: The system cancels the operation and leaves database data unchanged.

14.3. Special Requirements

Lecturers may only access their own profiles and update contact details. Deletion must strictly preserve data integrity.

Response time for creating, updating, or deleting lecturer profiles in the database must not exceed 15 seconds under standard test load conditions

14.4. Preconditions

The user is logged in.

14.5. Postconditions

Success: Lecturer profile is created, updated, or deleted in the database.

Failure: Lecturer records remain unchanged in the database; error message is displayed.

14.6. Extension Points

None

15. Assign Lecturer to Class Section

15.1. Summary

Allows the Administrator to assign a lecturer to a specific class section in a semester.

15.2. Flow of Events

15.2.1. Main Flow

1. The use case starts when the Administrator selects “Assign lecturer”.
2. The Administrator selects a semester and class section.
3. The system verifies section availability and displays the lecturer list.
4. The Administrator selects a lecturer and clicks Confirm.
5. The system checks the database for teaching schedule conflicts for the selected lecturer.
6. The system saves the assignment to the database.
7. Upon successful database commitment, the system displays a success message.

15.2.2. Alternative Flow

A1. At Step 5, if the lecturer has a schedule conflict with another section in the semester: the system rejects the assignment and displays “Assignment rejected: Schedule conflict detected for the selected lecturer with section [Class Code].”

A2. At Step 6, if saving fails: the system rolls back the transaction and displays “System error: Unable to save assignment. Transaction rolled back.”

15.3. Special Requirements

Only Administrators may assign lecturers.

15.4. Preconditions

The Administrator is logged in; the section and lecturer exist.

Response time for checking schedule conflicts and committing assignments to the database must not exceed 15 seconds under standard test load conditions

15.5. Postconditions

Success: The assignment is saved in the database; the lecturer gains authority to manage attendance and grades for the section.

Failure: Assignment status remains unchanged in the database; error message is displayed.

15.6. Extension Points

None

16. Manage Semesters

16.1. Summary

Allows the Administrator to create and configure semesters.

16.2. Flow of Events

16.2.1. Main Flow

1. The use case starts when the Administrator selects “Semester management”.
2. The system displays the semester list.
3. The Administrator chooses to add or edit a semester.
4. The Administrator enters: start/end dates, registration window, exam schedule timelines.
5. The system validates timelines (e.g., end date after start date, non-overlapping windows) and mandatory fields, then saves to the database.

6. Upon successful database commitment, the system displays a success message.

16.2.2. Alternative Flow

At Step 5, if dates are invalid (e.g., registration window outside semester dates), timelines overlap with another semester, or required fields are missing: The system rejects the request, performs rollback, and displays an error message: "Validation error: Invalid date configuration, conflicting timelines, or missing required fields."

16.3. Special Requirements

All academic activities, course registrations, and class sections must strictly link to a valid semester in the database.

Response time for validating timelines and saving semester data to the database must not exceed 15 seconds under standard test load conditions

16.4. Preconditions

The Administrator is logged in.

16.5. Postconditions

Success: Semester configuration is created or updated in the database.

Failure: Semester database records remain unchanged; error message is displayed.

16.6. Extension Points

None

17. Manage Faculties

17.1. Summary

Allows the Administrator to create, view, update, and delete faculties.

17.2. Flow of Events

17.2.1. Main Flow

1. The use case starts when the Administrator selects “Faculty management”.
2. The system queries the database and displays the faculty list.
3. The Administrator chooses to add, view, update, or delete.

4. Add/Update: The Administrator enters Faculty Code, Name, Email, Phone. The system validates unique Faculty Code and required fields, then saves to the database.
5. Delete: The system checks database constraints to verify if the faculty manages active courses or has enrolled students. If clear, the system prompts for confirmation. Upon confirmation, the system deletes the faculty from the database.
6. Upon successful database commitment, the system displays a success message with the result.

17.2.2. Alternative Flow

A1: At Step 4, if required fields are missing or the Faculty Code already exists in the database: The system rejects the request, performs rollback, and displays an error message: "Validation error: Missing mandatory data or duplicate Faculty Code."

A2: At Step 5, if the faculty manages active courses or has enrolled students linked in the database: The system rejects deletion, performs rollback, and displays an error message: "Deletion rejected: Faculty manages active courses or has enrolled students."

A3: At Step 5, if the Administrator cancels the confirmation dialog: The system cancels the operation and leaves database data unchanged.

17.3. Special Requirements

Faculty Code must be unique. Deletion must strictly be rejected if the faculty manages active courses or has enrolled students.

Response time for creating, updating, or deleting faculty data in the database must not exceed 15 seconds under standard test load conditions

17.4. Preconditions

The Administrator is logged in; for view/update/delete, the faculty record exists in the system.

17.5. Postconditions

Success: Faculty data is created, updated, or deleted in the database.

Failure: Faculty database records remain unchanged; error message is displayed.

17.6. Extension Points

None

18. Manage Courses

18.1. Summary

Allows the Administrator to create, view, update, and delete courses.

18.2. Flow of Events

18.2.1. Main Flow

1. The use case starts when the Administrator selects “Course management”.
2. The system displays the course list.
3. The Administrator chooses to add, view, update, or delete.
4. Add/update: The Administrator enters Course Code, Name, Credits, Managing Faculty, Prerequisites. The system validates unique Course Code, valid Managing Faculty, and required fields, then saves to the database.
5. Delete: The system checks database links to verify if the course is linked to ongoing class sections or historical academic records. If clear, the system prompts for confirmation. Upon confirmation, the system deletes the course from the database.
6. Upon successful database commitment, the system displays a success message.

18.2.2. Alternative Flow

A1: At Step 4, if mandatory fields are missing or the Course Code already exists in the database: The system rejects the request, performs rollback, and displays an error message: "Validation error: Missing mandatory data or duplicate Course Code."

A2: At Step 5, if confirmation is cancelled by the Administrator: The system cancels the operation and leaves database data unchanged.

18.3. Special Requirements

Course Code must be unique. Each course must be explicitly assigned to an existing managing faculty in the database.

Response time for validating and persisting course updates or deletions in the database must not exceed 15 seconds under standard test load conditions

18.4. Preconditions

The Administrator is logged in; the managing faculty exists; for view/update/delete the course exists.

18.5. Postconditions

Success: Course data is created, updated, or deleted in the database.

Failure: Course database records remain unchanged; error message is displayed.

18.6. Extension Points

None

19. Manage Class Sections

19.1. Summary

Allows the Administrator to manage class sections of a course within a semester.

19.2. Flow of Events

19.2.1. Main Flow

1. The use case starts when the Administrator selects “Class section management”.
2. The Administrator selects a semester; the system displays the section list.
3. The Administrator chooses to add, view, update, or delete.
4. Add/update: TThe Administrator enters Class Code, Course Code, Semester, Assigned Lecturer, Maximum Capacity, Room, and Class Schedule. The

system validates uniqueness, schedule conflicts, room capacity constraints, and automatically maintains Current Enrollment based on enrolled students, then saves to the database.

5. Delete: The system checks database constraints (e.g., student enrollments, existing grades). If clear, the system prompts for confirmation. Upon confirmation, the system deletes the section from the database.
6. Upon successful database commitment, the system displays a success message.

19.2.2. Alternative Flow

A1: At Step 4, if the assigned room or schedule conflicts with another class section in the database, or if enrollment exceeds Max Capacity: The system rejects the request, performs rollback, and displays an error message detailing the specific conflict: "Validation error: Conflict detected at Room [Room Name] / Schedule [Class Schedule] or Current Enrollment exceeds Maximum Capacity."

A2: At Step 4, if mandatory fields are missing or the Class Code already exists in the database: The system rejects the request, performs rollback, and displays an error message: "Validation error: Missing required fields or duplicate Class Code."

A3: At Step 5, if confirmation is cancelled by the Administrator: The system cancels the operation and leaves database data unchanged

19.3. Special Requirements

Each section must belong to one course and one semester. Class Code must be unique. Current Enrollment must never exceed Maximum Capacity. Conflict errors must explicitly name the conflicting room or schedule.

Response time for validating room/schedule conflicts and persisting section changes in the database must not exceed 15 seconds under standard test load conditions

19.4. Preconditions

The Administrator is logged in; the semester and course exist in the system; for view/update/delete, the section exists.

19.5. Postconditions

Success: Class section is created, updated, or deleted in the database.

Failure: Class section database records remain unchanged; error message is displayed.

19.6. Extension Points

None

CHAPTER 5: CLASS DIAGRAM

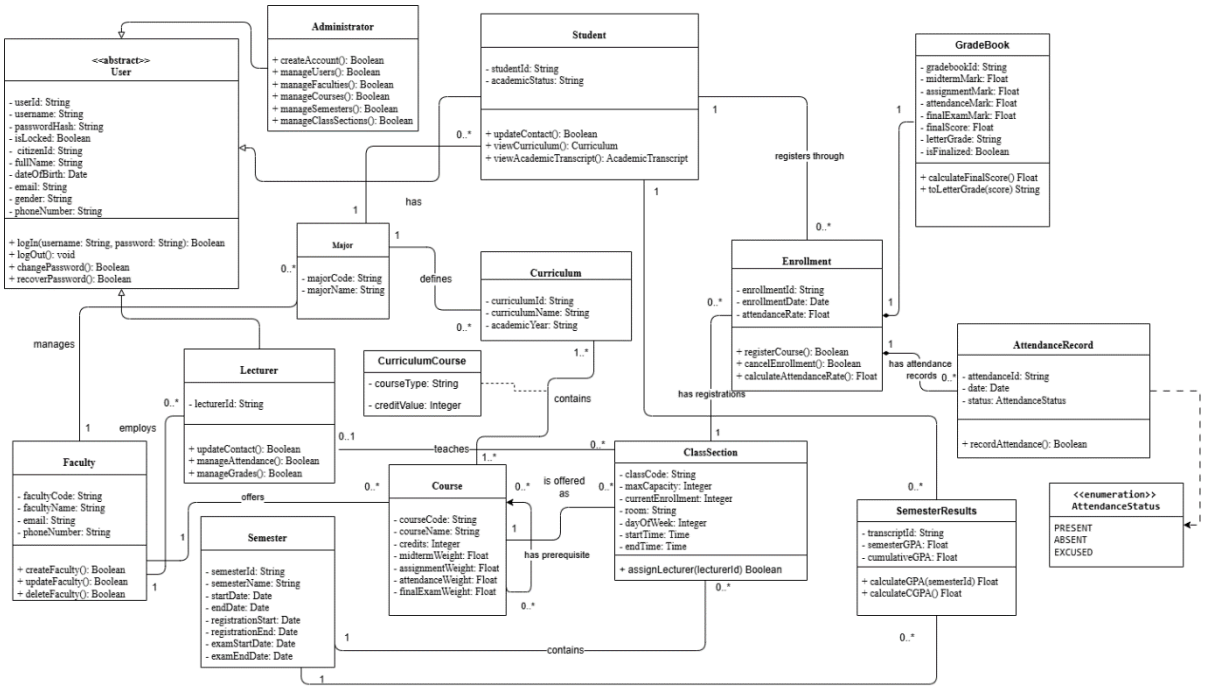


Image 4: Class Diagram

CHAPTER 6: DATA MODEL

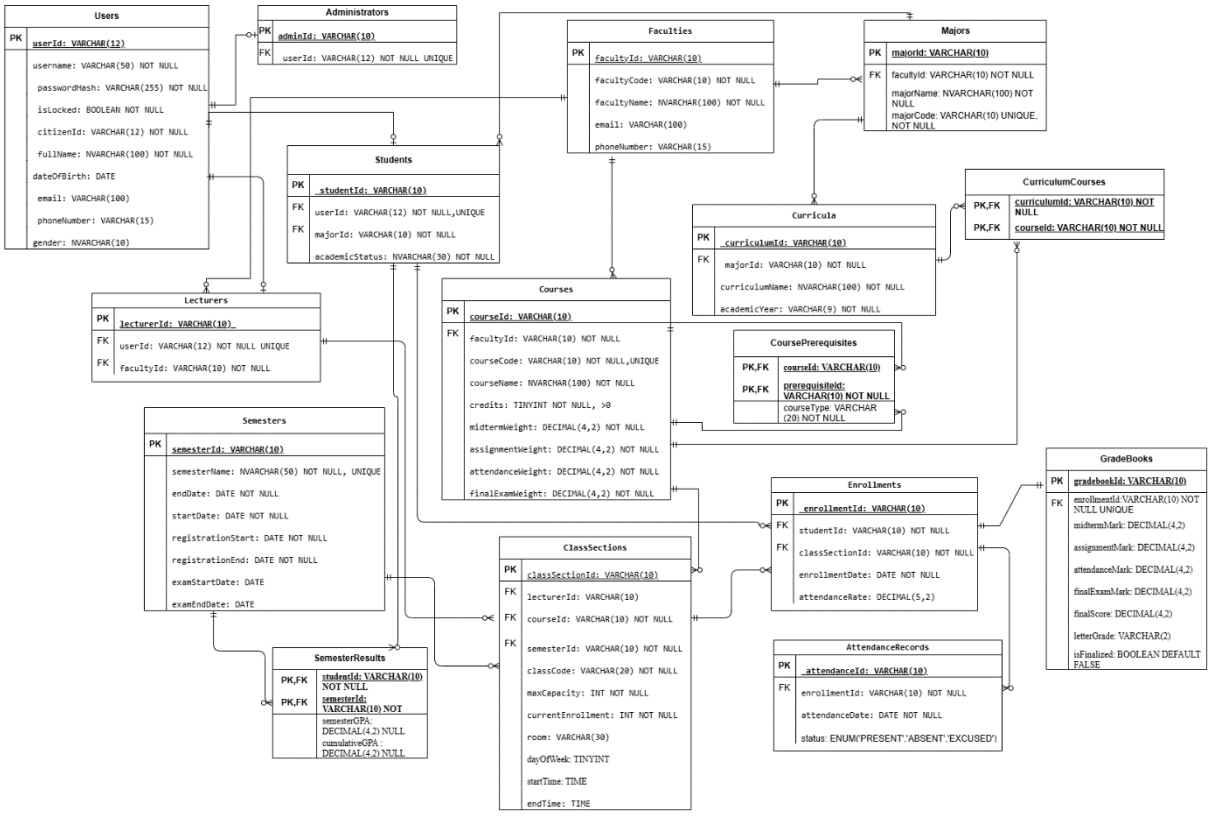


Image 5: Data Model

CHAPTER 7: INTERFACE DESIGN DESCRIPTION

1. Overall Interface Overview

The Student Management System (SMS) is a web-based application with a responsive graphical user interface (GUI). Figma is used to design the interfaces. In the next phase, the interface will be built with HTML, CSS, and JavaScript; Python will handle the main functions, and MySQL will store the data. The system will be deployed on a Windows 11 server

Users access the system through a web browser and work with visual forms, tables, dashboards, and navigation menus instead of commands. Based on user roles (admin, student, lecturer), the system follows role-based access control (RBAC) to provide separate menus with appropriate functions

2. Authentication Design

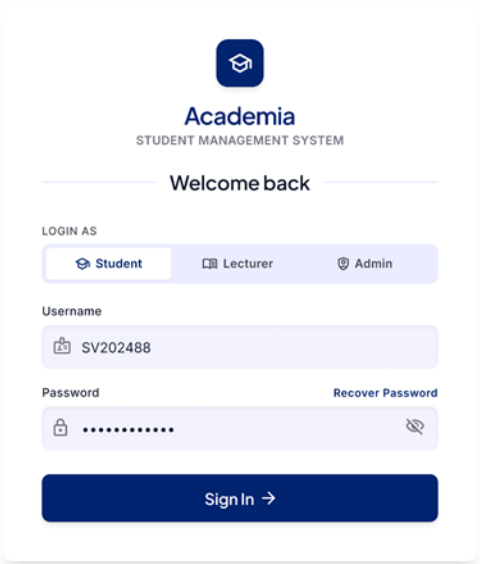


Image 6: Login Interface

The login page is the entry point for all users. It provides the following options:

- Select a role (student, lecturer, or admin), then log in using a username and password. The password is masked and can be shown or hidden. The selected role is only used to check the account. After a successful login, the system opens the interface of the role saved in the account.
- If the username or password is incorrect, the selected role does not match the account, or the account is locked, the system displays an appropriate error message without revealing sensitive account information. A role mismatch shows the same message as incorrect login information.
- Recover password through the registered email or phone number. After verifying the account, the user sets a new password directly on the screen.
- After 15 minutes of inactivity, the session ends and the user returns to the login page with the message “Session expired due to inactivity.”
- After logging in, users can log out from any screen.

3. Role-Based Menu Design

Each role is provided with a distinct set of functions

3.1. Admin Interface

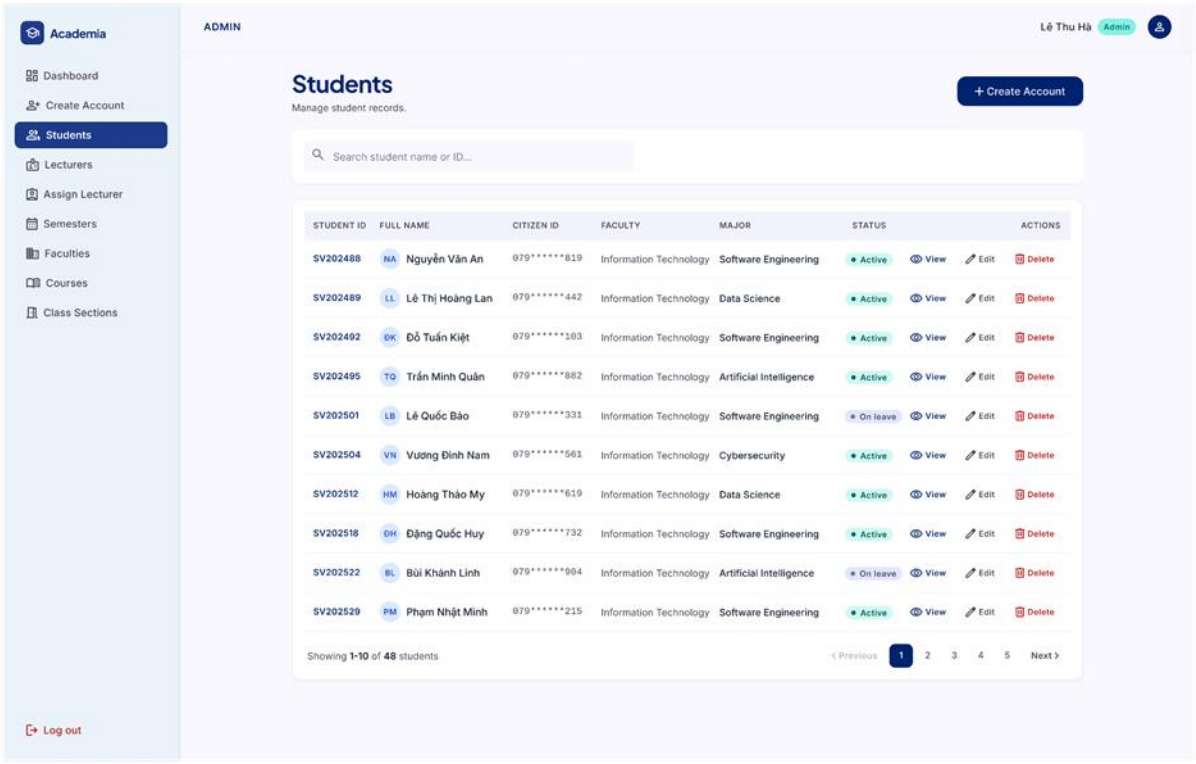


Image 7: Administrator Interface

The admin interface allows admins to manage user accounts and academic data. Its sidebar includes Dashboard, Create Account, Students, Lecturers, Assign Lecturer, Semesters, Faculties, Courses, Class Sections, and Log out. Main functions include:

- Creating student and lecturer accounts through Create Account. IDs are generated automatically. The “Create Account” button on the Students page opens the same form for the Student account type.
- Managing student records and lecturer profiles (view, create, update, delete) and searching by name or ID. Deleting a record requires a data check and confirmation.
- Assigning lecturers to class sections. The system checks for schedule conflicts.
- Managing Faculties, Courses, and Class Sections (view, create, update, delete).

- Managing Semesters (create, edit): dates, registration period, and exam schedule.
- Logging out of the system. Admins have no profile or change-password function.

3.2. Student Interface

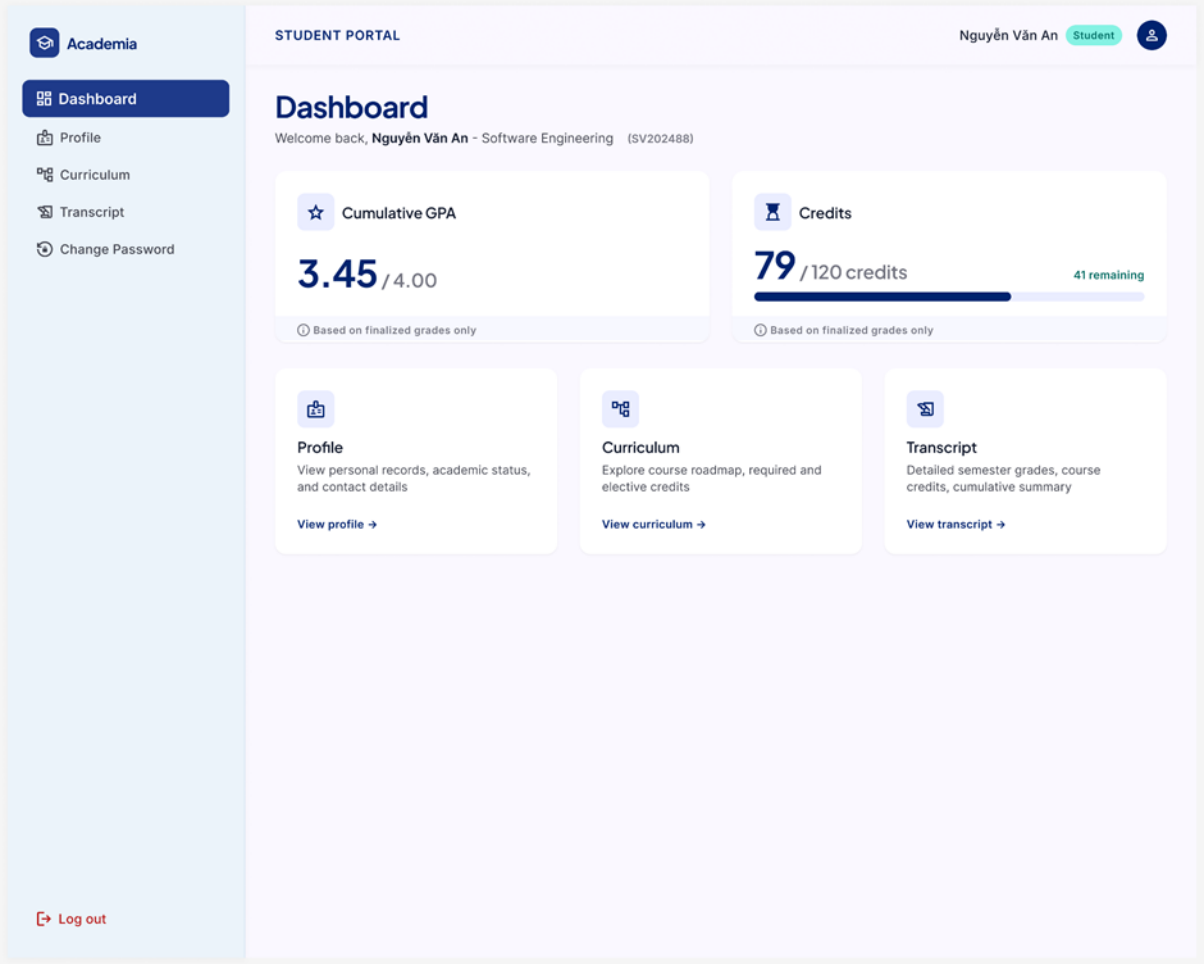


Image 8: Student Interface

The student interface supports academic tracking. Its sidebar includes Dashboard, Profile, Curriculum, Transcript, Change Password, and Log out. Students can perform the following actions:

- Viewing the dashboard, which displays the Cumulative GPA and the number of completed credits based on finalized grades only.
- Viewing their profile and updating contact information (email, phone number).
- Viewing the curriculum of their major, with each course labeled “Completed” or “Not Completed."
- Viewing the transcript, including finalized marks, letter grades, semester GPA, and cumulative GPA.
- Changing their password and logging out of the system

3.3. Lecturer Interface

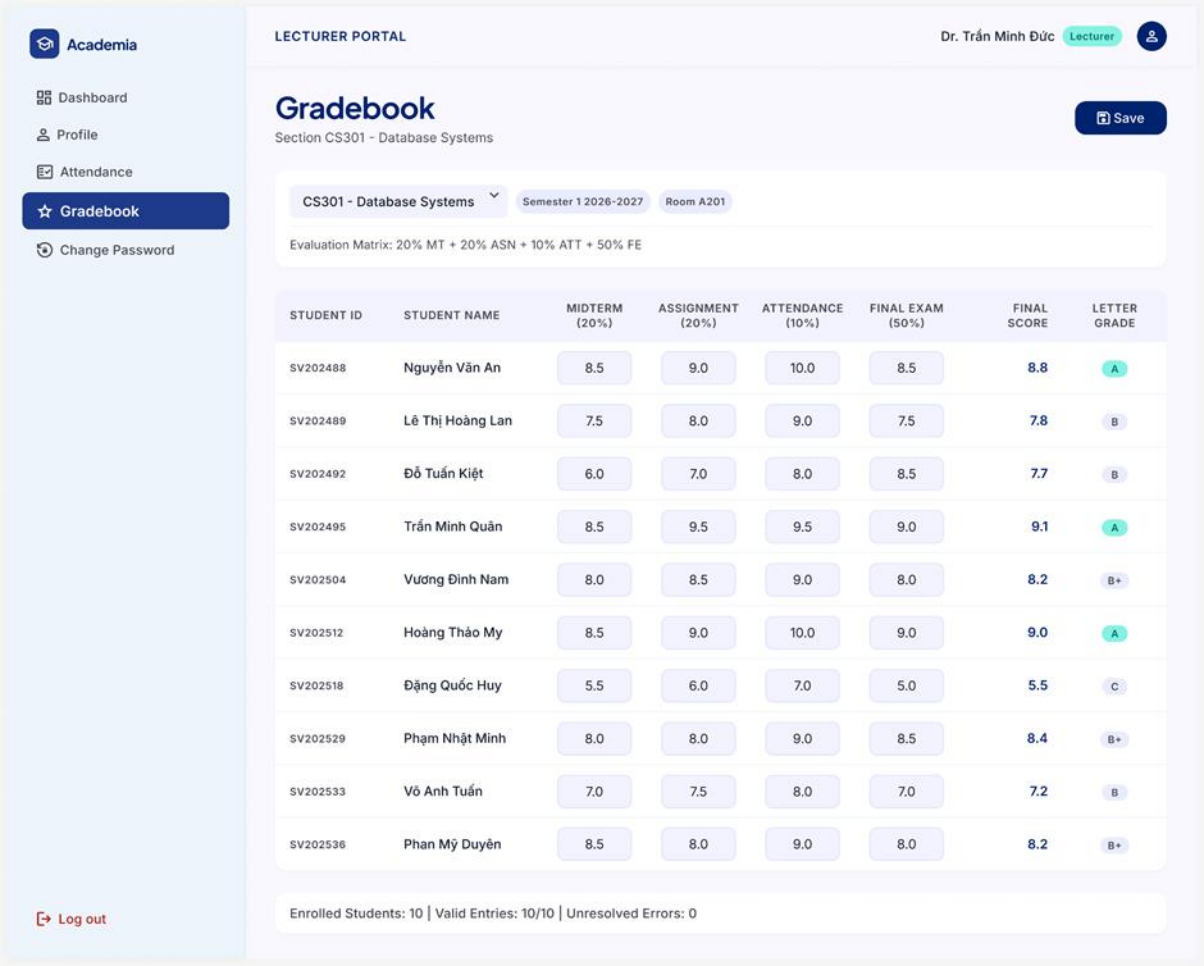


Image 9: Lecturer Interface

The lecturer interface provides teaching-related functions. Its sidebar includes Dashboard, Profile, Attendance, Gradebook, Change Password, and Log out. Lecturers can:

- View their profiles and update contact information, excluding their faculty.
- Take attendance (Present/Absent) for their assigned class sections in the active semester.
- Enter component marks (0.0-10.0) for their assigned class sections. The final grade and letter grade are calculated automatically.
- Change their passwords and log out of the system.

4. User Interaction and System Feedback

The system provides immediate and clear feedback after each user action.

- If an operation is successful, the system displays a success message. Important actions, such as changing a password or deleting a record, require confirmation.
- If an operation fails, the system displays an error message. The system prevents incomplete changes from being saved, and the original data remains unchanged.
- Error messages are clear and highlight the field causing the error, for example: “Validation error: [Field Name] is missing or already exists in the system.” or “Invalid grade format or out-of-range value at field [Field Name]. Grades must be between 0.0 and 10.0.”
- If a student or lecturer tries to change identification data or faculty, the system rejects the update. Only contact information can be changed.

5. Interface Design Rationale

A web-based GUI was selected for this system due to the following reasons:

- The non-functional requirements state that the interface must be easy to use, consistent, and responsive, with feedback after each action.
- Figma allows the group to design and prototype the interfaces before coding.
- Separate menus for each role reduce mistakes and support access control.
- A graphical interface can highlight the fields that cause errors.
- Tables and status labels display curriculum, transcripts, attendance, and gradebooks clearly.